BZAN554: Deep Learning

TUNING DEEP NEURAL NETWORKS

ON E-COMMERCE DATA

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KEY PERFORMANCE INSIGHTS

MAE Improvements Across Models:

- Models with moderate complexity **(2–3 layers)** performed best overall
- Learning rates around **0.0005–0.001** were the most stable
- **Early stopping** helped prevent overfitting and reduced training time
- **Embedding models** (SKU + category) performed better than basic models
- **Regularization** (dropout, batch norm, L2) improved model stability

BEST PRACTICAL MAE PROFILE

- 2–3 Hidden Layers
- 64–128 Neurons per Layer
- ReLU/ELU Family Activation
- Learning Rate ~0.001-0.0005
- Early Stopping Enabled

WHAT INCREASES RUN TIME

- **Smaller batch sizes (32–64)** → more updates per epoch → longer run time
- **More epochs** (without early stopping) → longer training
- **Lower learning rates** → require more epochs to converge
- **Larger/deeper networks** → increase computational cost
- **Embedding models** → slightly slower, but often improve performance

TUNING PARAMETERS

Parameters Tested Across Models:

- **Architecture:** 2-4 hidden layers, 32–640 neurons
- **Activations:** ReLU, ELU, Leaky ReLU, PReLU, Tanh, Sigmoid
- **Optimizers:** Adam, RMSprop, Adagrad, SGD (with Nesterov)
- **Learning rate:** ~0.0005 to 0.015
- **Batch size:** 32-256
- **Regularization:** dropout, batch norm, L2
- **Early stopping:** patience-based

TUNING MODEL OUTPUT

	hidden_layers	learning_rate	batch_size	epochs_ran	MAE	RMSE	training_time_sec
8	128->32	0.0010	64	6	12.78	30.27	403.62
10	128->32	0.0005	64	6	13.12	30.56	406.57
12	128->64	0.0010	64	5	13.13	30.41	359.25
14	128->64	0.0005	64	6	13.13	30.39	405.68
2	64->32	0.0005	64	7	13.20	30.64	406.83
3	64->32	0.0005	128	6	13.23	30.93	198.55
6	64->64	0.0005	64	6	13.24	30.62	381.09
4	64->64	0.0010	64	8	13.25	30.42	486.74
9	128->32	0.0010	128	7	13.25	30.58	265.64
11	128->32	0.0005	128	7	13.26	30.93	267.60
15	128->64	0.0005	128	8	13.27	30.67	258.79
7	64->64	0.0005	128	9	13.32	30.90	291.50
5	64->64	0.0010	128	5	13.41	30.53	141.13
0	64->32	0.0010	64	5	13.42	30.52	281.44
1	64->32	0.0010	128	7	13.45	30.86	203.04
13	128->64	0.0010	128	5	13.46	30.67	173.38

BEST PERFORMING MODEL

MAE

12.54

RUN TIME

126.12 sec (~ 2 minutes)

Model Configuration:

- **Hidden layers:** 2
- **Neurons:** 128→64
- **Activation:** ReLU
- **Optimizer:** Adam
- **Learning rate:** 0.001
- **Batch size:** 128
- **Epochs:** 20 with early stopping

CONTRIBUTIONS

Task:	Carys Van Atta	Katie Watts	Jazmin Elias	Olivia Helms	Tools Used:	Collaboration Methods:	Date Completed:
Designed Deep Neural Network Impemented in TensorFlow and Created Tuning Function	☑	☑	☑	☑	Python/VSCode, GitHub	Implemented a Deep Neural Network in TensorFlow with Tuning Function covering different batch sizes, number of hidden layers, etc.	3/30/2026
Gathered Experience on the Algorithm's Sensitivity to Tuning Parameters	☑	☑	☑	☑	Python/VSCode, GitHub	Tested and analyzed the effects on the model's performance after applying different tuning parameters including the number of hidden neurons, type of hidden activation function, etc.	3/30/2026 - 4/12/2026
Experimented With Architecture of Neural Network	☑	☑	☑	☑	GitHub, VSCode, Excel	Recorded results of each model run - documenting the results from each combination of tuning/training techniques explored.	3/30/2026 - 4/13/2026
Explored Different Training Methods	☑	☑	☑	☑	GitHub, Python, VSCode, Excel, GoogleDrive	Progress confirmed and results analyzed after each model run.	3/30/2026 - 4/13/2026
Compiled Useful Insights on the Sensitivity to Tuning Parameters	☑	☑	☑	☑	Excel, GoogleDrive	Analyzed each instance/output produced by our different models' results	4/09/2026 - 4/13/2026
Managed Zip File	☐	☑	☐	☐	Email, GoogleDrive	Ensured all materials were gathered and compressed into a ZipFile for submission.	4/13/2026
Create Presentation Deck	☑	☐	☐	☑	Canva, PowerPoint, Excel, iMessage, Email, FaceTime	Conbined all ideas discussed in meetings to create the final presentation slideshow deck.	4/09/2026 - 4/13/2026
Manage Run Book	☐	☐	☐	☑	GitHub, Excel, VSCode	Created and maintained Run Book, detailing all code used and it's purpose.	3/30/2026 - 4/13/2026

THANK YOU!